

Immersive Anatomy: Exploring Spatial Computing for Anatomy Education

Introduction

Immersive Anatomy is a visionOS educational application designed to improve the way students learn human anatomy through immersive three-dimensional visualization, guided learning pathways, and interactive assessment. The project explores how emerging spatial computing technologies can be used to address challenges in anatomy education by transforming passive learning experiences into interactive and immersive educational journeys.

Human anatomy is inherently three-dimensional. However, most educational resources still rely heavily on textbooks, diagrams, presentations, and static illustrations. While these resources are valuable, they often require learners to mentally reconstruct spatial relationships between structures. This process can be difficult, particularly for students who are encountering anatomy for the first time.

The goal of this project was to investigate how immersive technologies can help learners better understand anatomical structures by combining realistic 3D visualization, structured educational content, and knowledge reinforcement within a single learning environment.

Research and Problem Validation

Before beginning development, I conducted preliminary research to understand how anatomy is currently taught and learned.

I spoke with medical students and practicing doctors associated with Tashkent Medical University. During these discussions, I explored their experiences learning anatomy, the educational resources they relied on, and the challenges they encountered throughout their studies.

Several recurring themes emerged.

Students consistently described anatomy as one of the most difficult subjects during the early stages of medical education. Although textbooks contain extensive information, understanding the spatial relationships between structures often requires significant mental effort. Students frequently reported needing to examine multiple diagrams from different angles before they could develop a complete mental model of an organ.

Practicing doctors emphasized that anatomy forms the foundation of medical education and clinical practice. They highlighted the importance of developing strong spatial understanding early in the learning process, as this knowledge supports later study in physiology, pathology, surgery, and clinical decision-making.

These conversations motivated me to investigate how immersive technologies could support anatomy education. Rather than simply creating a 3D model viewer, I wanted to design a complete educational experience that guides learners from observation to understanding and finally to knowledge reinforcement.

The insights gained during this research phase directly influenced the structure and design of Immersive Anatomy.

Problem Statement

Traditional anatomy education relies heavily on two-dimensional resources. Although these resources effectively communicate information, they often fail to naturally convey depth, scale, orientation, and spatial relationships.

As a result, learners must perform the additional cognitive task of translating two-dimensional information into three-dimensional understanding.

This project addresses the following question:

How can spatial computing be used to create a more intuitive, engaging, and effective anatomy learning experience while maintaining educational rigor?

Proposed Solution

Immersive Anatomy addresses this challenge through a structured educational framework built around four complementary learning modes:

Explore

Explore Mode serves as the learner's first contact with an organ.

In this mode, users can observe realistic three-dimensional anatomical models within an immersive environment. The goal is to encourage curiosity and initial discovery without overwhelming the learner with information.

Explore Mode helps learners answer:

"What am I looking at?"

Labels

Labels Mode introduces anatomical terminology through contextual identification.

Instead of immediately presenting detailed explanations, users are guided through the major structures of an organ using carefully placed labels that identify key anatomical components.

This stage acts as a bridge between visual recognition and conceptual understanding.

Labels Mode helps learners answer:

"What is this structure called?"

Learn

Learn Mode is the core educational component of the application.

Each organ follows a structured learning journey consisting of multiple chapters organized into categories such as:

- Overview
- Anatomy
- Functionality
- Real-Life Relevance
- Clinical Insight
- Summary

Each chapter combines professionally designed medical illustrations with concise educational explanations, key facts, and contextual information.

This structure guides learners from foundational concepts toward deeper understanding.

Learn Mode helps learners answer:

"How does this organ work?"

Quiz

Quiz Mode reinforces learning through active recall and assessment.

Questions are directly connected to the educational content presented in Learn Mode, allowing users to verify their understanding and apply newly acquired knowledge.

This stage transforms passive learning into active learning.

Quiz Mode helps learners answer:

"Can I remember and apply what I learned?"

Educational Framework

The learning experience is intentionally organized around a progressive educational pathway:

Explore → Labels → Learn → Quiz

This mirrors a natural learning process:

Observe → Identify → Understand → Apply

Rather than presenting educational content as disconnected screens or features, each mode builds upon the previous one and supports a specific stage of learning.

This framework was one of the key design decisions of the project and contributes significantly to both its educational value and overall user experience.

Technical Frameworks

RealityKit

RealityKit serves as the foundation for all three-dimensional content within the application.

It is responsible for:

- Loading USDZ anatomical models
- Rendering organs in real time
- Managing scene composition
- Positioning educational content around anatomical structures
- Creating immersive visual presentations

RealityKit was selected because it provides a robust framework for spatial experiences while integrating seamlessly with visionOS.

visionOS

The application leverages visionOS immersive spaces to create a dedicated learning environment.

Instead of restricting educational content to conventional windows, the application combines three-dimensional anatomy, educational information, and assessment tools within a shared spatial context.

This allows users to remain focused on the organ being studied while accessing supporting educational material.

SwiftUI

SwiftUI was used to construct the user interface and manage application state.

SwiftUI powers:

- Organ selection
- Educational navigation
- Learn Mode chapter presentation
- Labels Mode
- Quiz interfaces
- Progress tracking
- Dynamic content updates

The declarative architecture allows the application to remain maintainable while supporting complex educational workflows.

Spatial Computing Principles

Although the application does not rely heavily on advanced gesture recognition or hand-tracking interactions, it demonstrates several important principles of spatial computing.

The project uses space as part of the learning experience rather than merely as a container for content.

Key spatial computing concepts employed include:

- Three-dimensional educational visualization
- Immersive learning environments
- Contextual information placement
- Spatial relationship awareness
- Integration of educational content around a central learning object

The anatomical models are not decorative assets. They function as the primary educational objects around which the learning experience is organized.

This approach allows learners to develop stronger mental models of anatomy than would be possible through traditional two-dimensional resources alone.

System Architecture

The application was designed around modular and scalable principles.

The architecture separates educational content from presentation logic, allowing new organs, lessons, and assessments to be introduced without significant architectural changes.

The system consists of:

- Organ Visualization Layer
- Labels System
- Educational Content Layer
- Learn Mode Architecture
- Quiz Engine
- User Interface Layer

This separation improves maintainability while supporting future expansion.

The current implementation includes:

- Heart
- Brain
- Lungs
- Kidneys

with an architecture capable of supporting additional organs in future versions.

Technical Proficiency

The project demonstrates technical proficiency through:

- Effective use of RealityKit for educational visualization
- Integration of immersive spaces within visionOS
- State-driven architecture using SwiftUI
- Modular educational content organization
- Reusable learning chapter system
- Scalable quiz architecture
- Structured learning progression
- Consistent design system implementation

The project was developed entirely by me, including:

- Application architecture
 - RealityKit integration
 - Immersive environment design
 - Labels Mode implementation
 - Learn Mode architecture
 - Educational content system
 - Quiz engine
 - User interface implementation
 - Learning progression design
 - Educational research and validation
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Elegance of the Solution

I believe the elegance of the solution lies in the way technology and pedagogy are aligned.

Many educational applications separate visualization, theory, and assessment into unrelated experiences. Immersive Anatomy instead integrates these elements into a single educational journey.

The application follows a progression that mirrors natural learning behavior:

Explore → Labels → Learn → Quiz

This sequence corresponds directly to:

Observe → Identify → Understand → Apply

By organizing the experience around this educational framework, the application reduces cognitive overload while creating a more intuitive learning process.

The result is a solution that is both technically sophisticated and educationally meaningful.

Practical Value

Immersive Anatomy demonstrates how spatial computing can be applied to a real educational challenge.

Rather than using immersive technology as a novelty, the project leverages it to improve understanding of complex three-dimensional structures.

The combination of:

- Immersive visualization
- Anatomical identification
- Structured learning content
- Interactive assessment

creates a comprehensive educational experience that supports both engagement and comprehension.

Beyond anatomy education, this approach has potential applications in other scientific and technical domains where spatial understanding plays an important role.

Future Development

Future iterations of the project could explore:

- Additional anatomical systems
- More advanced assessment methods
- Multi-user collaborative learning experiences
- Personalized learning pathways
- Enhanced spatial interactions
- Clinical case-based educational modules

The current architecture was intentionally designed to support these future extensions.

Conclusion

Immersive Anatomy explores how spatial computing can contribute to more effective anatomy education.

Inspired by conversations with medical students and practicing doctors, the project investigates an alternative approach to learning anatomy by combining immersive visualization, structured educational content, and active assessment.

Using RealityKit, visionOS, and SwiftUI, the application transforms anatomy learning from a passive reading experience into an interactive spatial experience.

More importantly, the project demonstrates how emerging technologies can be applied thoughtfully to meaningful educational challenges. By combining research, educational design, and technical implementation, Immersive Anatomy seeks to make learning anatomy more intuitive, engaging, and effective.

The project reflects my interest in spatial computing, educational technology, and the creation of experiences that use technology not only to inform, but to improve understanding.